Swiss Paraplegic Research

Exercise: R Markdown

Workshop – Introduction to R

1 Creating an R Markdown report

In this exercise, you are supposed to play around with R Mardown to gain experience. For this purpose, you will write a small mock report about the smoking behavior of youths in Finland. For comparison, you find an example solution of the exercise in the **exampleSolution** folder.

To get started, open a new R Markdown file in R Studio. Once opened, insert the `output: bookdown::word_document2` line in the YAML header to use this output. Add a bit of text and a first section header and then try to compile the file into a Word file. You can add a first R chunk to your file in which you set some default chunk options and load the needed libraries.

1.1 Getting the data

We will use data from the Global Youth Tobacco Survey 2012 of which we will only use the data from Finland (to make it easier to handle). To get the data, go to the official webpage (link)[1] and accept the terms and conditions to download the microdata as a .csv file. Add an R chunk to your file in which you read-in the downloaded .csv file. Take a look at the data to make sure the import worked correctly.

1.2 Pre-processing the data

For this analysis, we will only use four items from the questionnaire. Table 1 shows the four wanted items and their meaning. For more detailed information about the items and their response options check the **Data Dictionary** tab (choose **FINLAND_12**) on the questoinnaire's webpage. Additionally, Table 1 shows the pre-processing that you need to conduct on these variables. Try to perform the described pre-processing steps to prepare the data for analysis.

Table 1: Selected variables and their required pre-processing.

Item	Meaning	Required pre-processing
CR2	Sex	Turn the variable into a factor with the levels female and male (check the webpage to know which number corresponds to which sex). Rename the variable appropriately.

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[1] Link address: https://extranet.who.int/ncdsmicrodata/index.php/catalog/532/get-microdata

Table 1 – *Continued from previous page*

Item	Meaning	Required pre-processing
CR40	Intention to use tobacco in the next 12 months	The intention to use tobacco was answered on a four level scale. Check the webpage for the exact meaning of the answer options. Turn the variable into a factor with only two levels, where the first level ("no") corresponds to negative answers (1, 2) and the the second level ("yes") corresponds to positive answers (3, 4). Rename the variable appropriately.
OR15, OR16	How often father (OR15) or mother (OR16) is seen smoking	The two items correspond to how often the father or mother is seen smoking. Again, check the answer options online. As can be seen online, each question can be answered on a four level scale: Answers 2 and 3 indicate that the parent is seen smoking, while answers 1 and 4 deny that. We want to combine those two variables into one factor with two levels, which indicate whether either the mother or the father is ever seen smoking or not (`parents_smoke`: no, yes). Rename the variable appropriately.

1.3 Introduction section

Add a small introduction section. Describe in the section that we are interested in investigating the relationship between the presence of smoking parents (newly created binary variable) and the intention to smoke in the next 12 months. State where the data was obtained from and add a click-able link to the questionnaire's webpage (find out with the internet how to add links in a Markdown file).

Additionally, we want to show that this questionnaire has been used for other investigations of the smoking behavior of youths by referencing the following study: *Prevalence of e-cigarette use and its associated factors among youths aged 12 to 16 years in 68 countries and territories: global youth tobacco survey* (Sun et al. 2022). Cite this paper in your text using a BibTex file.

Finally, add some descriptive statistics to your introduction section like the total number of observations in the data set and the number of female and male participants.

1.4 Statistical analysis

Create a 2x2 frequency table which shows the numbers of participants who do/don't intend to smoke when parents are/aren't seen smoking (creating such tables was shown in the *Data Visualization* lecture, **hint**: `table()`). Add a statistical analysis section to your report in which you show this table.

Afterwards, conduct a Chi-squared test to statistically test the nullhypothesis that there is no relation between the two variables. A Chi-squared test can be conducted with the `chisq.test` function which takes the created 2x2 table as an input. Report the results in your report, showing the test statistic and corresponding p-value.

1.5 Data visualization I

Visualize the 2x2 frequency table as a grouped barplot and display the figure in your report. Make a cross-reference to the figure in your text.

1.6 Data visualization II

Additionally, we want to visualize the proportions of participants who do/don't intend to smoke with two pie charts. One pie chart for participants with smoking parents and one pie chart for participants without smoking parents. Show those two plots next to each other in your report.